

Methods Appendix — Store Quality Intelligence Program

Program overview. SentinelQA measures store quality through structured mystery-shop visits. Each visit produces a 100-point overall score from SentinelQA's proprietary weighted rubric. January–May 2026 visits were conducted by SentinelQA's trained professional shopper panel on a fixed schedule (four visits per store per month, balanced across dayparts). On June 1, 2026 the program transitioned to the Community Shopper platform, in which vetted app users complete visits at times and locations of their choosing, scored under rubric v2.

Data dictionary — visits.csv

Column	Description
visit_id	Unique visit identifier
store_id	Store identifier (S001–S120)
visit_date	Date of visit (2026)
daypart	morning / midday / evening
shopper_id	Shopper identifier
shopper_type	professional app_volunteer
wait_time_min	Minutes from entry to order completion
order_accurate	1 if order fulfilled exactly as placed
score_speed	Speed-of-service component (0–100)
score_service	Service-quality component (0–100)
score_cleanliness	Cleanliness component (0–100)
overall_score	Overall quality index (0–100), weighted composite per SentinelQA rubric

Data dictionary — store_master.csv

Column	Description
store_id	Store identifier
region	North / South / East / West
near_university	TRUE if within one mile of a university campus (24 stores)
open_date	Store opening date
closure_start / closure_end	Scheduled renovation closure window, if any (store closed to customers)

Statistical methodology

Historical visit-level standard deviation: $\sigma = 6.5$ points (2019–2025 program history; used as known σ in all control computations). Program standard: 16 visits per store per quarter. Out-of-control threshold: chain mean $- 2 \times (\sigma / \sqrt{16})$. Confidence intervals: mean $\pm 1.96 \times s / \sqrt{n}$ at 95%. Scores are calibrated where appropriate. SentinelQA's rubric, calibration procedures, and shopper vetting standards are proprietary.